

Typhoid Disease Prediction Using Multiple Machine Learning Models and Explainable AI

Abstract— Typhoid fever is a devastating disease that kills millions of people annually as a result of delayed diagnosis and economic underdevelopment. Nevertheless, this research offers a thorough comparison of eight Machine Learning (ML) models constructed especially for categorical classification in an environmental scenario. The evaluated models contain traditional classifiers, hybrid ensemble model and sophisticated boosting models including support vector machine, k-nearest neighbor, decision tree, random forest, naïve bayes, logistic regression, (random forest+XGBoost), gradient boosting and XGBoost. When all ML models are used, the hybrid ensemble model yields the highest accuracy, at 98.78%. This indicate that hybrid ensemble model is more effective than standalone models. Additionally, we employ Explainable AI (XAI) which incorporates SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), to visualise risk factors and provide more accurate disease predictions. This research proposes a strong, interpretable and adaptable classification framework suitable for environmental modelling and broader ML applications, emphasizing the importance of comparing algorithms and using variety of features for efficient predictive modelling. It also uses a simplified web-based framework for interactive, real-time prediction and visualisation.

Keywords—Machine Learning, Hybrid Ensemble Model, Explainable AI, Typhoid Fever Disease.

I. INTRODUCTION

Health is the foundation of all happiness and sustainable development. But maintaining a good health is a critical challenge as infectious and hazardous diseases continue to affect millions of people every year. From these disorders, typhoid fever remains a global public health concern. Even in the twenty first century, typhoid fever is still a serious public health issue in developing nations [1]. Due to patient hospitalization, typhoid fever also has an economic and societal consequences [2]. According to World Health Organization (WHO) data, 9 million people worldwide contract typhoid each year and 0.11 million people pass away from the disease [3]. Inadequate hygiene practices, contaminated water and poor sanitation all contribute to spread of typhoid fever. In Bangladesh, typhoid is particularly prevalent in crowded cities, unsanitary conditions and contaminated drinking water sources.

Typhoid fever is caused by a bacterium known as *Salmonella typhi* (s.typhi). Consuming tainted food, drink or beverages can transmit S.typhi. Once in the intestines and bloodstream, the germs can travel other areas of the body [4]. Although the exact nature of this interaction is unknown, it may involve fimbriae, S.typhi possesses 12 fimbrial operons of the chaperone usher assembly class, none of which are specific to S.typhi, including tcf (Typhi colonization factor) [5]. However, s.typhi possesses a unique collection of fused operons; the varied fimbriae family may have been selected

for by the host immune response [6]. S.typhi also has a type IVB pilus operon (pil) and pilS :: $km^r(pil^-)$ mutants are less able to adhere to and infiltrate intestinal epithelial cells than wild type(pil^+) S.typhi [7]. Typhoid disease spreads across poor sanitation, unsafe water and inadequate hygiene practices.

Bangladesh is predicted as a high-risk country for typhoid fever due to its rapid urbanization, inadequate sanitation etc. Bangladesh is severely affected with an annual incidence rate of 252 per lakh affected individuals [8]. Moreover, in many parts of the country, the danger of typhoid transmission is also increased by unforeseen flooding and contaminated water, lack of proper use, financial burdens such as treatment expanses and lack of diagnostic resources. Even though Upazila Health Complexes (UHCs) are the initial primary health referral facilities, there are other shortcomings, such as lack of electricity and maintenance backup [9].

The availability of intelligent systems is increasing in the modern world [10]. In recent times, Artificial Intelligence (AI) plays a crucial role in medical sector such as enhancing disease diagnosis, treatment planning and healthcare management. By analyzing patient information, AI based system can recognize sophisticated configuration which is difficult to detect by using traditional system. Some specific AI methods that are significant in medical area named as a rule-based intelligent system, gives a collection of if-then rules in healthcare, which operate as a decision support system. In the medical industry, intelligent system are gradually being supplanted by AI- based autonomous methods that require very little human participation [11]. Explainable AI (XAI) like SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations) is a branch of AI. Transparency, interpretability and explainability of the results are all included in XAI, “The ability to explain or to present in understandable terms to a human” is how Doshi-Velez and Kim define interpretable in the context of XAI [12][13].

The main contribution of this paper are :

- Rather than employing a typical machine learning model, the study uses an hybrid ensemble model (XGBoost+ Random Forest) for improved accuracy and early prediction.
- XAI methods, such as SHAP and LIME, are used to make model decisions more transparent and trustworthy. Making educated decisions in the field of human healthcare is facilitated by the use of interpretable machine learning.

II. METHODOLOGY

This segment expresses the strategy used to expand a Machine Learning (ML) classification system that employs demographic and serological features extracted from the

dataset to identify typhoid in human patients. This also includes feature selection, data preprocessing, model training using a variety of machine learning and comprehension using SHAP and LIME procedures.

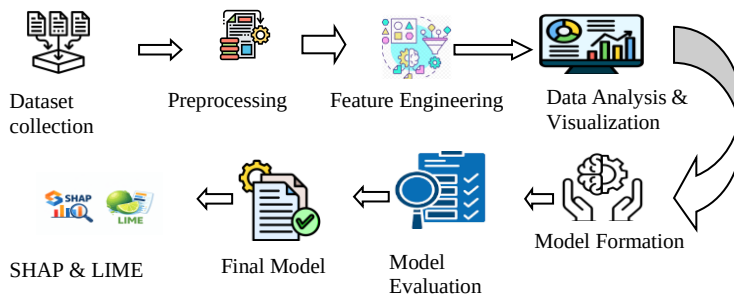


Fig-1: Block diagram for proposed model

A. Data collection and preprocessing:

The Mendeley TyphoDx-BD: A Clinically Valid Dataset for Machine Learning – Based Typhoid and Rickettsial Disease Prediction in Bangladesh archive dataset [14] includes demographic and serological features related to typhoid disease generality. The dataset contains important features for instances TH (typhoid H antigen), TO (typhoid O antigen), AH (antigen H), OX9, Rickettsia_suspect and paratyphoid_A and the target feature denotes typhoid Negative, Minimum and positive. The dataset was prepared for supervised classification by normalizing continues variables and encoding categorical ones after missing values were removed. 17 pertinent elements, including demographic and serological factors, were found in 1107 entries using exploratory data analysis.

B. Feature selectinon and feature Engineering:

There are demographic, laboratory test and diagnosis variables in the dataset. Among the important characteristics looked at are:

Demographic variables: Name, age and gender are all covered in this category of variable.

Laboratory test variables: It includes TH, TO, BH, AH, OX2, OX9, OXK, A and M.

Diagnosis variables: It contains Rickettsia-Suspect, Acute-typhoid, ParatyphoidA, ParatyphoidB.

Target Variables: The intended variable for the outcome is the last feature, typhoid.

Feature engineering is crucial for turning the raw data that has been gathered into useful input features. It is also employed to improve the ML models interpretability and predictive power, as well as increased computational effectiveness.

C. Exploratory Data Analysis:

The relationship between meteorological factors and the incidence of typhoid sickness was examined using three different kinds of visualizations: Feature correlation with

typhoid, histogram of numerical features and correlation matrix heatmap.

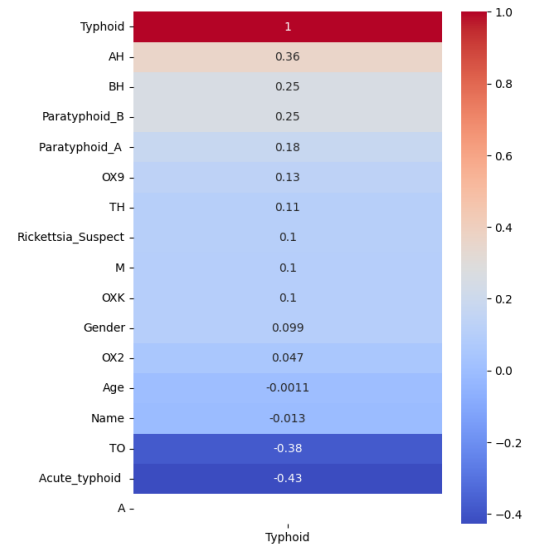


Fig-2: Feature correlation with typhoid

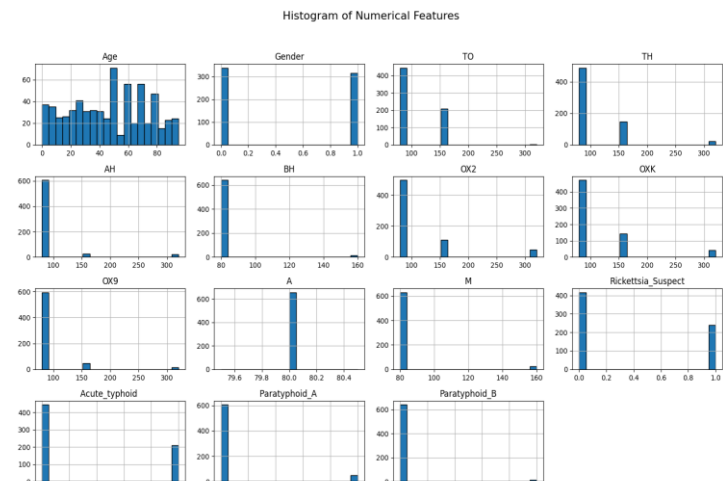


Fig-3: Histogram of numerical features

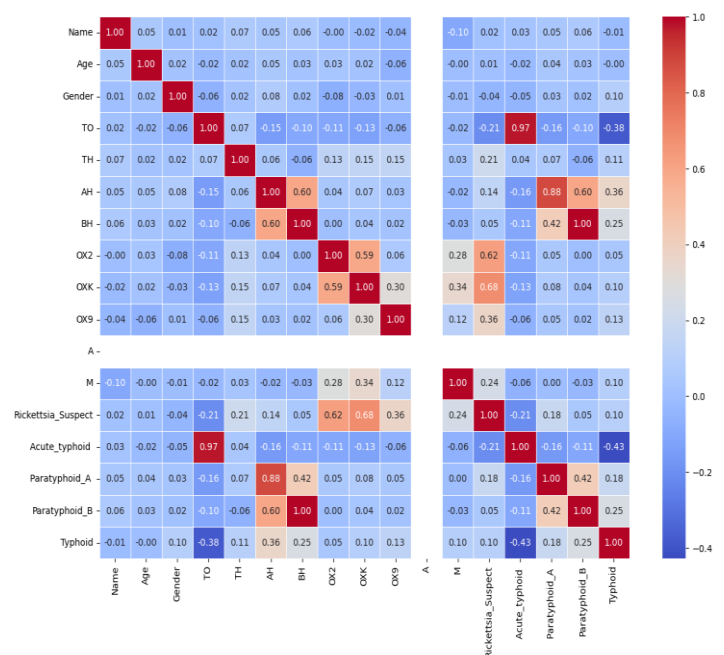


Fig-4: Correlation matrix heatmap

Fig-2 shows the feature correlation map, which identifies the key characteristics and presents the data pattern for several feature types that are connected to typhoid disease. Strong features affect prediction performance and redundant or less informative features suggest that improved feature selection is needed. It also shows the direction and relationship between the variables. Fig-3 displays a histogram of numerical features that are utilized to exploit data imbalance and the data distribution of numerical features. Moreover, by detecting statistical distribution and density variation, the histogram is employed to determine the frequency with which certain numerical values occur in the dataset. Model robustness and stability are increased as a result of this visualization's support for improved data preparation and normalization technique. Fig-4 displays a correlation matrix heatmap, which is intended to identify multicollinearity and visually depict the characteristics and relationship with typhoid. The correlation heatmap uses color intensity variations to show the pairwise correlation coefficient, identify highly colinear variables that could introduce redundancy and improve the classification's stability and interpretability.

D. Model Classification:

Support Vector Machine (SVM): A support vector machine(SVM) develops hyperplane or multiple hyperplanes which exist in either high-dimensional or infinite-dimensional spaces to perform classification and regression and various other functions. The hyperplane which achieves optimal separation between classes exists at maximum distance from the closest training sample because classifiers show decreased generalization error which increasing separation distance. The fundamental equation is:

$$\min_{W,b} \frac{1}{2} ||w||^2 + C \sum_{i=1}^n \varepsilon_i \quad [15]$$

K-Nearest Neighbors (KNN): KNN uses a distance measure like Euclidean distance to classify a data point based on the most prevalent label among its K nearest neighbors. Furthermore, KNN classifies new data by comparing it with existing samples and maintains all training samples for classification. Appropriate selection clearly contributes to improve model accuracy and reduces misclassification.

$$D(x_i, x_j) = \sqrt{\sum_{m=1}^M (x_{im} - x_{jm})^2} \quad [16]$$

Decision Tree (DT): To improve class purity, decision trees use measures like Gini impurity to partition data based on feature thresholds. This method effectively handles both numerical and categorical data while organizing the dataset recursively to yield the most information gain. The necessary equation is:

$$\text{Gini} = 1 - \sum_{i=1}^c p_i^2 \quad [17]$$

Random Forest (RF): One machine learning method that uses meta-learning is the random forest algorithm [18]. For a given set of inputs, it establishes an overall classification using a variety of random tree classifications. This method decreases overfitting, increases the model's capacity for generalization and yields a feature importance score.

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_T(x)\} \quad [19]$$

XGBoost: Complex datasets can be handled with XGBoost, while minimizes overfitting and provides exceptional accuracy while handling missing values. While maintaining a high computational speed, XGBoost effectively manages large datasets with missing values and captures intricate feature interactions. The basic formula is:

$$\text{Obj} = \sum_{i=1}^n l(y_i, \hat{y}_j) + \sum_{k=1}^K \Omega(f_k) \quad [20]$$

Logistic Regression: Logistic Regression is useful for binary classification and is used to solve classification problems. It serves as a baseline model as well. Moreover, this model performs extremely well on basic classification tasks and clearly illustrates the influence of each feature on the prediction.

Gradient Boosting: Gradient Boosting builds additive models in a progressive manner, with each new model resolving the shortcomings of its predecessors. In this context, this model optimizes gradient descent technique to minimize prediction error and employs random subsampling strategy to prevent model overfitting.

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x) \quad [21]$$

Naïve Bayes: Because it adheres to the Bayes theorem, Naïve Bayes is a fast and efficient algorithm used for probability based classification. For problem including text classification and high-dimensional datasets, the Naïve Bayes model also performs well and efficiently. Additionally, it only needs a little quantity of data in the training set to function consistently and predictably.

Hybrid Ensemble Model (XGBoost+Random Forest): The hybrid ensemble model is used for stable and balanced predictions and provides robust and reliable forecasts. Furthermore, it outperforms individual algorithms on difficult data, improves generalization on unknown data and offers high accuracy. This type of combination lowers errors and produces more accurate predictions.

E. Model Evaluation :

The classification models performance was assessed using a number of measures for a comprehensive examination. The main metric used to evaluate accuracy

was the ratio of correctly identified instances to the total number of samples.

Moreover, a confusion matrix was examined to shed light on particular mistake kinds and the classification outcomes in terms of true positive, true negative, false positive and false negative in order to improve the study of model performance.

F. Final Model:

When compared to typical machine learning models, the suggested hybrid ensemble model offers more accurate results as well as robust and consistent performance.

G. Model Insights:

We employed XAI with SHAP and LIME to classify typhoid disease prediction in order to increase hybrid ensemble model's intelligibility. SHAP offers a thorough view on feature relevance by presenting a reliable and theoretically sound way for evaluating each input features contribution to model categorization [22]. However, the goal of LIME is to create interpretable models that correctly explain individual predictions, making complicated models easier for humans to understand [23]. By modifying these AI methods, we may more accurately anticipate typhoid disease and enhance our honesty and practicality in real-world applications.

III. RESULTS AND DISCUSSION:

The observations of the study revealed important information on the diagnosis and clinical symptoms of typhoid in selected patients. The accuracy comparison of several machine learning models for the provided dataset is displayed in Fig-5.

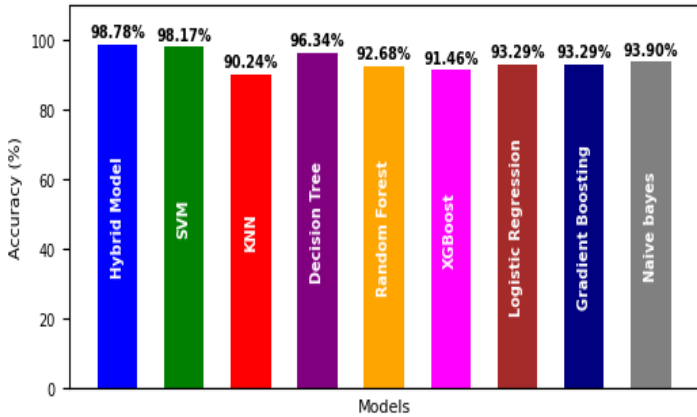


Fig-5: Model Accuracy Comparison

Compared to traditional machine learning models, the hybrid ensemble model (XGBoost+Random Forest) achieves the highest accuracy (98.78%), as seen in the above picture. The ensemble model is the best approach to this classification problem because of its outstanding performance. On the other hand, traditional ML model provides the accuracy slightly behind: SVM (98.17%), KNN(90.24%), Decision Tree(96.34%), Random Forest(92.68%), XGBoost(91.46%),

Logistic Regression(93.29%), Gradient Boosting(93.29%) and Naïve Bayes (90.30%).

TABLE I: Comparison of typhoid disease study

Ref.no	Best Model	XAI
[24]	XGBoost=97.87%	No
[25]	ANN=22%	No
[10]	SVM=96%	NO
[26]	XGBoost=96.1%	NO
[27]	Random forest=77.5%	No
[28]	Multi boost decision stump=94%	No
Our model	Hybrid ensemble model=98.78%	Yes

TABLE II: Classification report of based performance model

Class	Precision	Recall	F1- Score	Support
0	0.99	0.99	0.99	70.0
1	0.99	1.0	0.99	82.0
2	1.0	0.92	0.96	12.0
Accuracy	0.99	0.97	0.99	0.99
Macro Avg	0.99	0.99	0.98	164
Weight Avg	0.99	0.99	0.99	164

A classification report, which includes precision, recall, F1-score and support is crucial since it assures a detailed understanding of the model's effectiveness. The model's great ability to accurately categorise the feature is demonstrated by the above Table, which shows that it's total accuracy achieved 99%. Because the model stored 99% precision, 99% recall and 99% F1-score, class 0 demonstrates a balanced and dependable performance. Additionally, class 1 indicates that the model was able to identify nearly real characteristics and class 2 achieved 100% precision, 92% recall and 96% F-1 score. For prediction tasks, micro average and weight average guarantee model stability and overall efficiency.

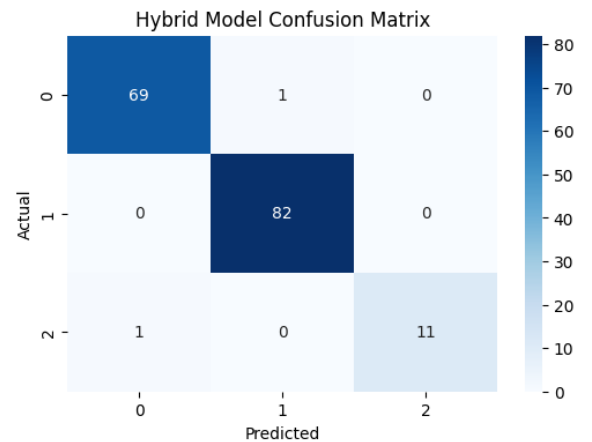


Fig-6: Confusion Matrix of Based performance model

The confusion matrix for the optimal model performance is shown in figure-6 above, which also offers a good classification capability for the three classes. Class-0 correctly predicts 69 cases while misclassifying 1 instance. Additionally, class-1 correctly predicts 82 instances with no misclassified instances, while class-2 correctly predicts 11 instances with one misclassified instance.

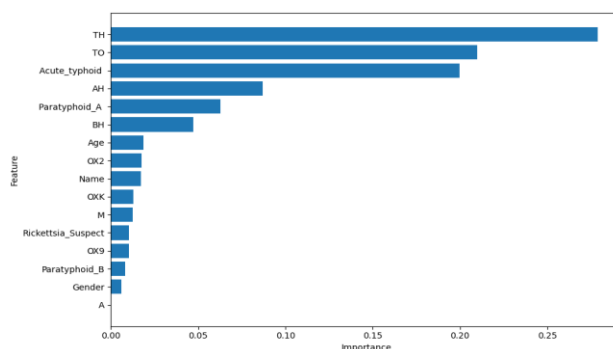


Fig-7: Feature importance of based performance model

The hybrid feature relevance analysis in Fig-7 highlighted the crucial factors influencing the classification procedure. From the above observation, it is seen that TH feature has the highest importance score while TO is the second most influential parameter and Acute_typhoid achieves the third highest priority. Additionally, other feature provides moderate importance. However, TO serves as an early disease indication, acute_typhoid recognizes the intensity of typhoid symptoms and TH antigen indicates the active immune response. Since clinical validation is crucial and this kind of integration technique enhances model transparency, this kind of explanation for a specific disease is significant and facilitating early detection of typhoid disease.

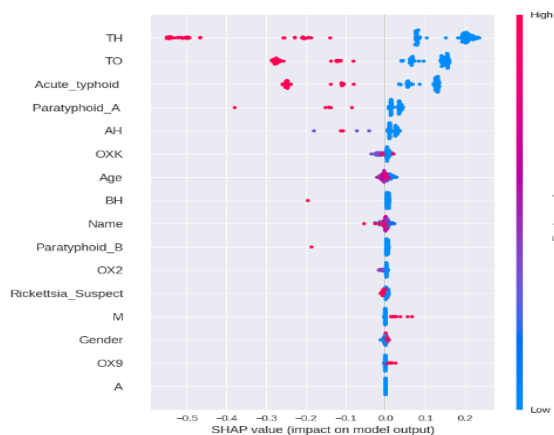


Fig-8: SHAP summary plot for feature importance

Fig-8 displays the SHAP summary plot, which demonstrates which features are important for disease prediction and which characteristics provide a global interpretation of feature relevance. According to the figure, the most important characteristics for classification performance are TH, TO and Acute_typhoid, whereas gender and rickettsia_suspect have less of an impact on the prediction outcome. It is evident that TH and TO are crucial in differentiating typhoid disease. The SHAP visualisation demonstrates that assessing each input

features contribution to model categorization is a trustworthy and theoretically valid method.

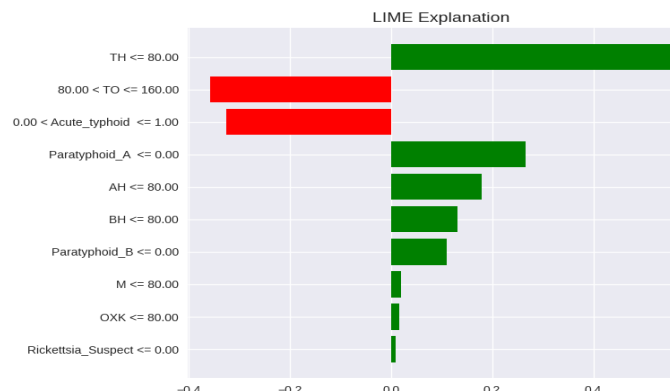


Fig-9: LIME Explanation plot

As seen in Fig-9, we can see that, The LIME analysis provided comprehensive details about the interpretability of individual predictions. When it involves to typhoid prediction, TH contribute significantly negative and TO contribute significantly positive. Additionally, it is shown that paratyphoid_a, AH, BH and paratyphoid_b contribute moderately positively but in a lower amount. Ultimately, it explains that LIME's goal is to develop interpretable models that accurately explain individual predictions, simplifying complex models for human interpretation.

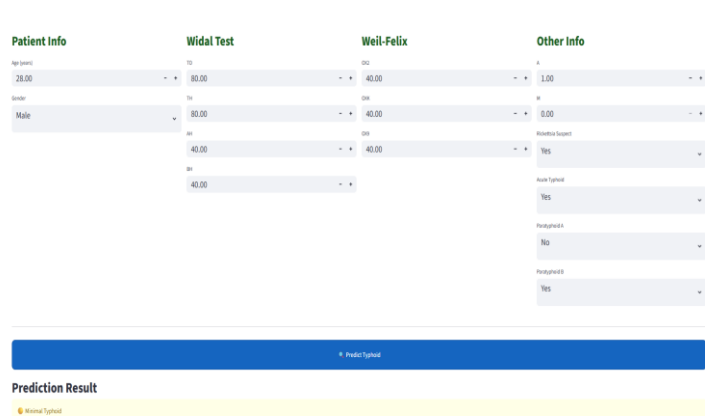


Fig-10: Streamlit Web Based Framework

A simplified web based framework is offered by fig-10, which is used to increase convenience, flexibility and accessibility. Furthermore, it makes accurate predictions in real time without requiring sophisticated technical knowledge. When inputs are received, the system uses a trained model to turn the data into information and produce an accurate prediction result that aids in determining the disease's stage. Ultimately, the trained hybrid prediction model is integrated with a graphical user interface using streamlit.

IV. CONCLUSION

This study uses clinical data to investigate the effectiveness of multiple ML model for typhoid disease forecasting. To, investigate their ability to predict, a total of eight unique classification models were implemented. The models comprise both sophisticated hybrid ensemble techniques

(Random Forest+XGBoost) and conventional algorithms (SVM, KNN, Random Forest, Decision Tree, Naïve Bayes, XGBoost, Gradient Boosting). These models are used to compare and demonstrate accuracy. However, the traditional ML model performs moderately, while the hybrid model offers the maximum accuracy. By providing a dependable and theoretically valid method for assessing each input feature's contribution to model categorization, SHAP provides a comprehensive perspective on feature significance. LIME, on the other hand, aims to develop interpretable models that accurately explain individual predictions, simplifying complex models for human comprehension. Additionally, we make use of the free source streamlit web framework, which enables us to rapidly and easily develop interactive visualisation dashboards. Despite many obstacles, this study offers insightful information for further research: The analysis is restricted to a single geographic dataset due to data availability and no doctor recommendation system is in place. The model will be updated for future work by incorporating a multi-regional dataset and most importantly, Rickettsial Disease Prediction in Bangladesh", Mendeley Data, V1, an automated doctor referral module to improve its practical applicability and ultimately cure this type of deadly disease

REFERENCES

- [1] A. Oguntimilehin, O. B. Abiola, and K. A. Olatunji, "Computer Aided Diagnostic Systems for Managing Typhoid Fever: A Review of Diagnosis Techniques," *International Journal of Computer Applications*, vol. 126, no. 6, pp. 24–29, Sep. 2015, doi: 10.5120/ijca2015906071.
- [2] WHO (2007), "Taking Stock: Health Workers Shortages and the Response to AIDS", www.who.int/health_systems/.../TTR. Retrieved 04/04/14.
- [3] World Health Organization, "Typhoid," WHO, [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/typhoid>. [Accessed: Mar. 30, 2023].
- [4] S. A. Fatumo, E. Adetiba, and J. O. Onolapo, "Implementation of XpertMaltyp: An expert system for medical diagnosis of the complication of malaria and typhoid," *IOSR Journal of Computer Engineering*, vol. 8, no. 5, pp. 34–40, Jan.–Feb. 2013. [Online]. Available: <http://www.iosrjournals.org>
- [5] S. M. Townsend, N. E. Kramer, R. Edwards, et al., "Salmonella enterica serovar Typhi possesses a unique repertoire of fimbrial gene sequences," *Infection and Immunity*, vol. 69, pp. 2894–2901, 2001.
- [6] T. L. Nicholson and A. J. Bäuml, "Salmonella enterica serotype Typhimurium elicits cross-immunity against a *Salmonella enterica* serotype Enteritidis strain expressing LP fimbriae from the lac promoter," *Infection and Immunity*, vol. 69, pp. 204–212, 2001.
- [7] X. L. Zhang, I. S. Tsui, C. M. Yip, et al., "Salmonella enterica serovar Typhi uses type IVB pili to enter human intestinal epithelial cells," *Infection and Immunity*, vol. 68, pp. 3067–3073, 2000.
- [8] D. P. Forster and K. Leder, "Typhoid fever in travellers: Estimating the risk of acquisition by country," *Journal of Travel Medicine*, vol. 28, no. 8, Art. no. taab150, 2021.
- [9] Y. Salman, H. Asim, N. Hashmi, Z. Islam, M. Y. Essar, and M. A. Haque, "Typhoid in Bangladesh: Challenges, efforts, and recommendations," *Annals of Medicine and Surgery*, vol. 80, Art. no. 104261, 2022.
- [10] A. Oguntimilehin, A. O. Adetunmbi, and O. B. Abiola, "A machine learning approach to clinical diagnosis of typhoid fever," *International Journal of Computer and Information Technology*, vol. 2, no. 4, pp. 1–6, Jul. 2013.
- [11] C. C. Lee, "Fuzzy logic in control systems: Fuzzy logic controller. I," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 20, no. 2, pp. 404–418, Mar./Apr. 1990.
- [12] P. Linardatos, V. Papastefanopoulos, and S. Kotsiantis, "Explainable AI: A review of machine learning interpretability methods," *Entropy*, vol. 23, pp. 1–45, 2021.
- [13] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," arXiv:1702.08608, Feb. 28, 2017. [Online]. Available: <https://arxiv.org/abs/1702.08608>
- [14] M. Shahriar, R. A. Nayeem, S. M. A. Al Muhib, and M. H. I. Bijoy, "TyphoDx-BD: A clinically validated dataset for machine learning-based typhoid and rickettsial disease prediction in Bangladesh," *Mendeley Data*, vol. 1, 2025, doi: 10.17632/m9pnvv2vpv.1.
- [15] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [16] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Transactions on Information Theory*, vol. 13, no. 1, pp. 21–27, Jan. 1967, doi: 10.1109/TIT.1967.1053964.
- [17] L. Breiman, J. Friedman, R. Olshen, and C. Stone, *Classification and Regression Trees*. Belmont, CA, USA: Wadsworth International Group, 1984.
- [18] F. Livingston, "Implementation of Breiman's random forest machine learning algorithm," ECE591Q Machine Learning Journal Paper, pp. 1–13, 2005.
- [19] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, Oct. 2001.
- [20] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD '16)*, pp. 785–794, 2016.
- [21] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [22] S. M. Lundberg and S.-I. Lee, "Consistent feature attribution for tree ensembles," arXiv:1706.06060, 2017. [Online]. Available: <https://arxiv.org/abs/1706.06060>.
- [23] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD '16)*, pp. 1135–1144, 2016.
- [24] M. A. Bhuiyan, S. S. Rad, F. T. Johora, A. Islam, M. I. Hossain, and A. A. Khan, "Prediction of typhoid using machine learning and ANN prior to clinical test," in *Proc. Int. Conf. Computer Communication and Informatics (ICCCI)*, pp. 1–7, 2023.
- [25] O. A. Abisoye, I. Doughlas, and B. O. Abisoye, "A machine learning based approach for the management of typhoid and malaria infection," in *Proc. 11th Int. Multi-Conf. ICT Application (AICTTRA)*, 2017.
- [26] P. O. Odion and E. O. Ogbonnia, "Web-based diagnosis of typhoid and malaria using machine learning," *Nigerian Defence Academy Journal of Military Science and Interdisciplinary Studies*, vol. 1, no. 2, 2024.
- [27] M. Hussain, M. A. Cifci, T. Sehar, S. Nabi, O. Cheikhrouhou, H. Maqsood, M. Ibrahim, and F. Mohammad, "Machine learning based efficient prediction of positive cases of waterborne diseases," *BMC Medical Informatics and Decision Making*, vol. 23, no. 1, p. 11, 2023.
- [28] K. T. Igulu, T. P. Singh, T. C. Olayinka, O. H. Onyijen, E. Onatuyeh, and W. Nwankwo, "An intelligent approach to cost-effective management of typhoid fever in children," in *Proc. 10th Int. Conf. Signal Processing and Communication (ICSC)*, pp. 265–271, 2025.